MPTE Workshop

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NANOG98

Agenda

- MPTE Overview
  - Basics
  - Provisioning
    - RSVP
    - BGP
- Workshop Overview

Why MultiPath Traffic Engineering (MPTE)

Networks are increasingly built with inherent multipathing attributes

- ECMP, while present, does not address all networking challenges e.g. Bandwidth Management

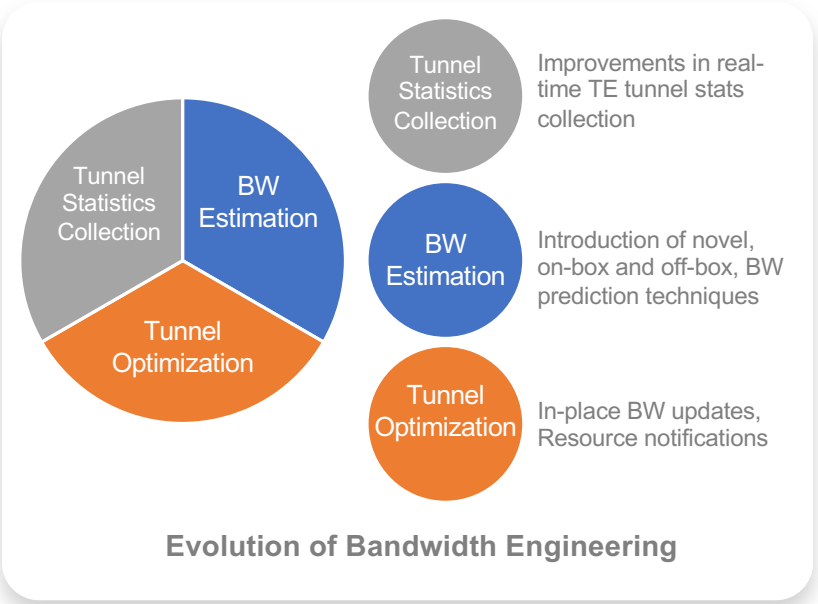
Next Evolution in Bandwidth Engineering

- Bandwidth engineering (e.g. Auto-bandwidth) has arguably been the best tool in the TE toolkit
- Steady investment/evolution of auto-bandwidth for more than a decade

MPTE is the next step in that evolution

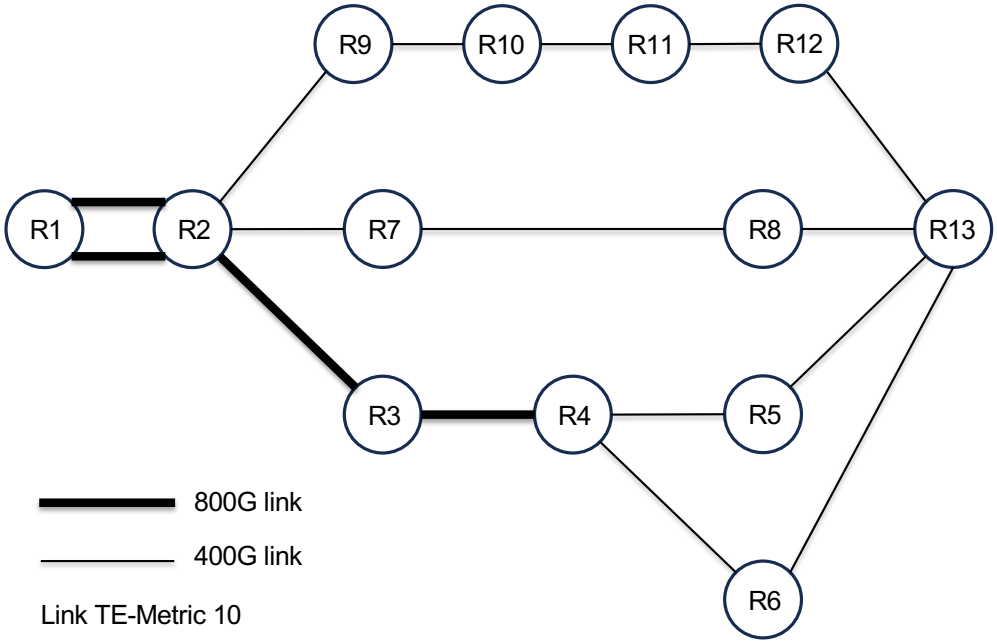
Adds TE to multi-path networks

Adds multi-path to TE networks



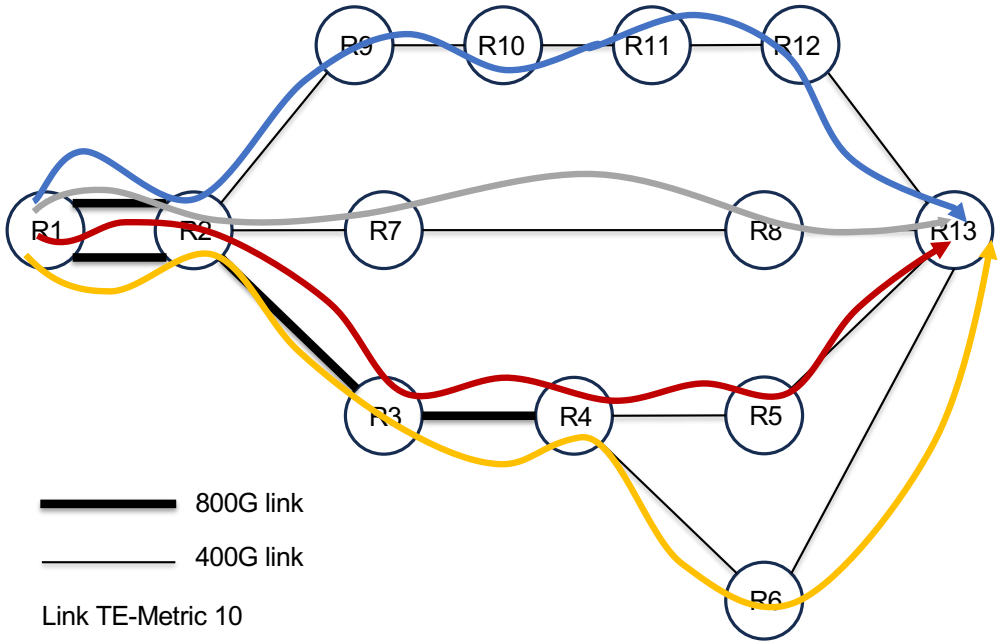
Evolution of Bandwidth Engineering

Example Topology 1



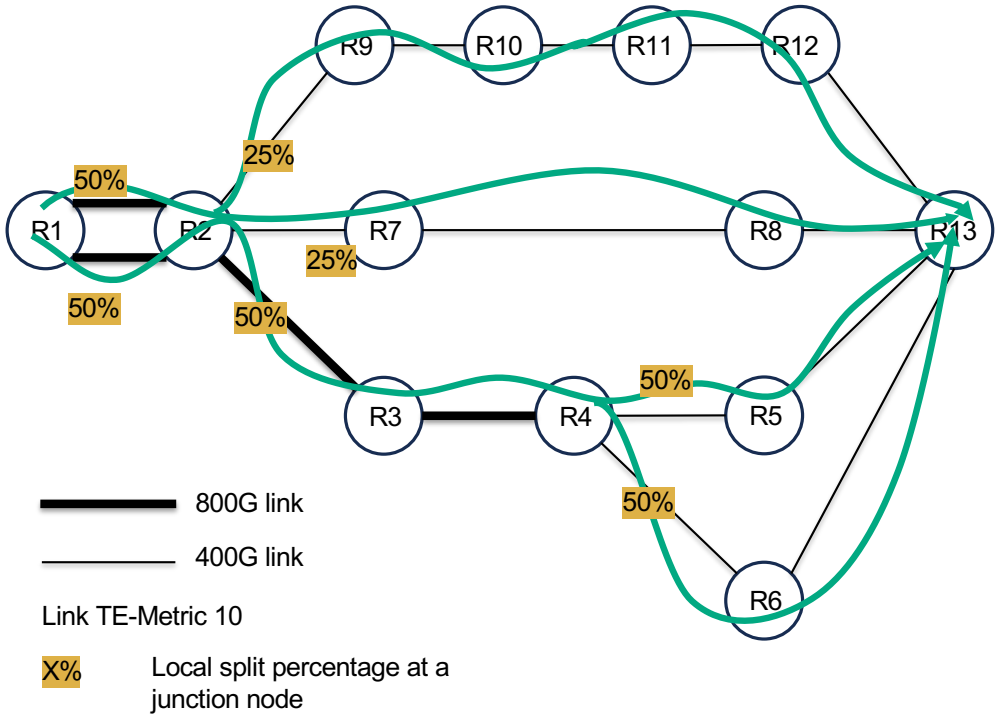
Maximum flow throughput from R1 to R13: 1.6 Tbps

Multipathing using Conventional TE [1]



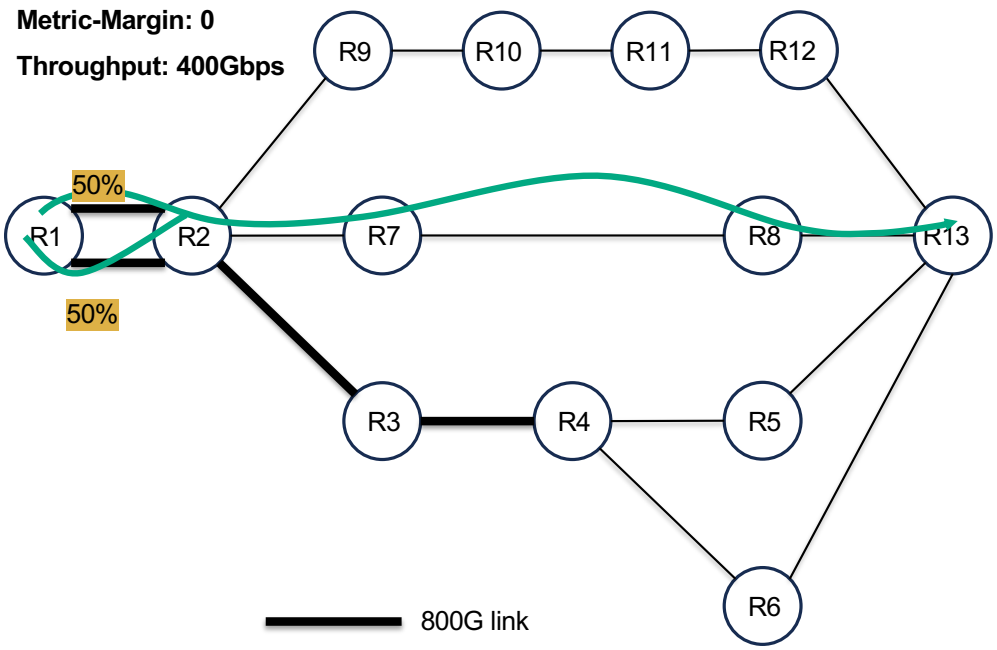
- To achieve 1.6 Tbps throughput between R1 and R13 --
  - Spawn and maintain 4 individual TE Tunnels
    - Manually created (or)
    - Dynamically created using TE++ (Container Tunnels)

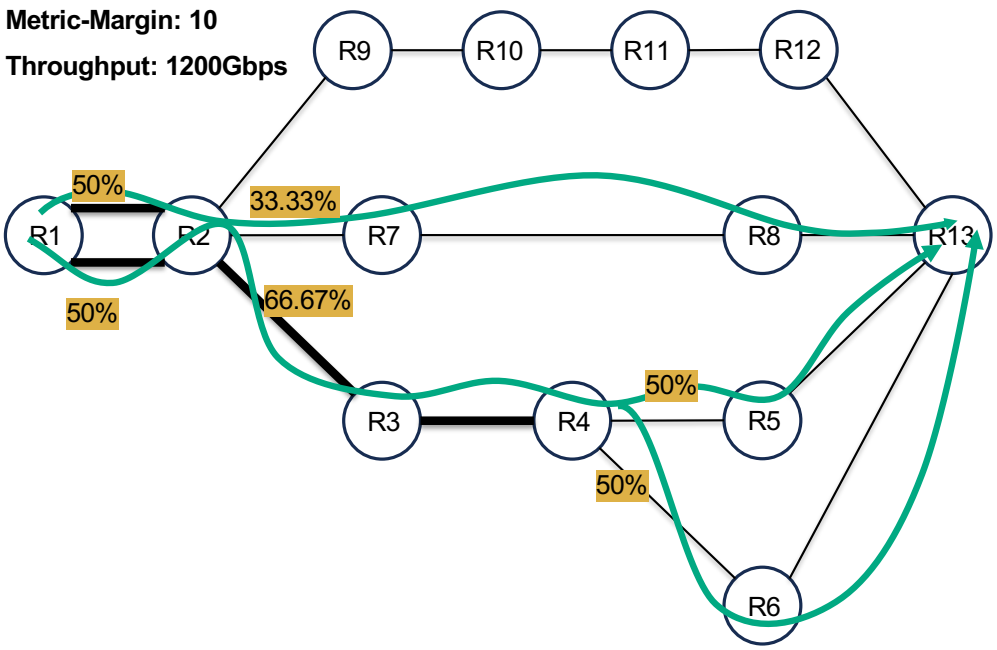
Multipathing using Directed Acyclic Graph (DAG) based MPTE [1]



- To achieve 1.6 Tbps throughput between R1 and R13 --
  - Spawn and maintain one MPTE DAG Tunnel
  - DAG can branch with an optimal split ratio at each junction and recombine as needed

Adjusting the MPTE DAG Span using Slack

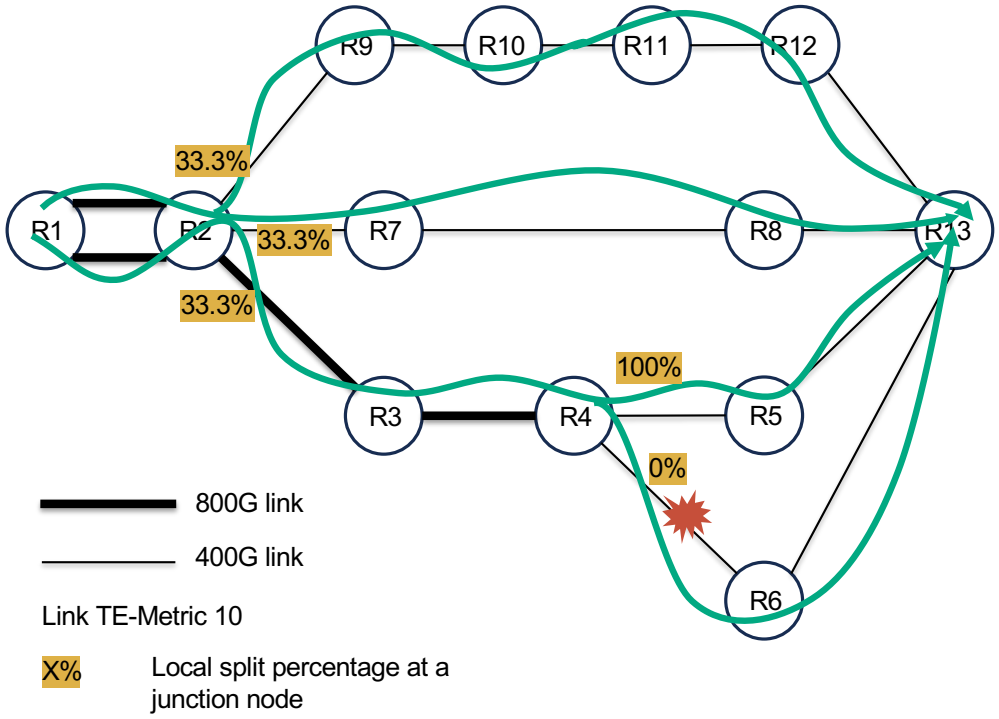




- To achieve 1.6 Tbps throughput between R1 and R13 –
  - Use metric-margin 20 (or default infinite)

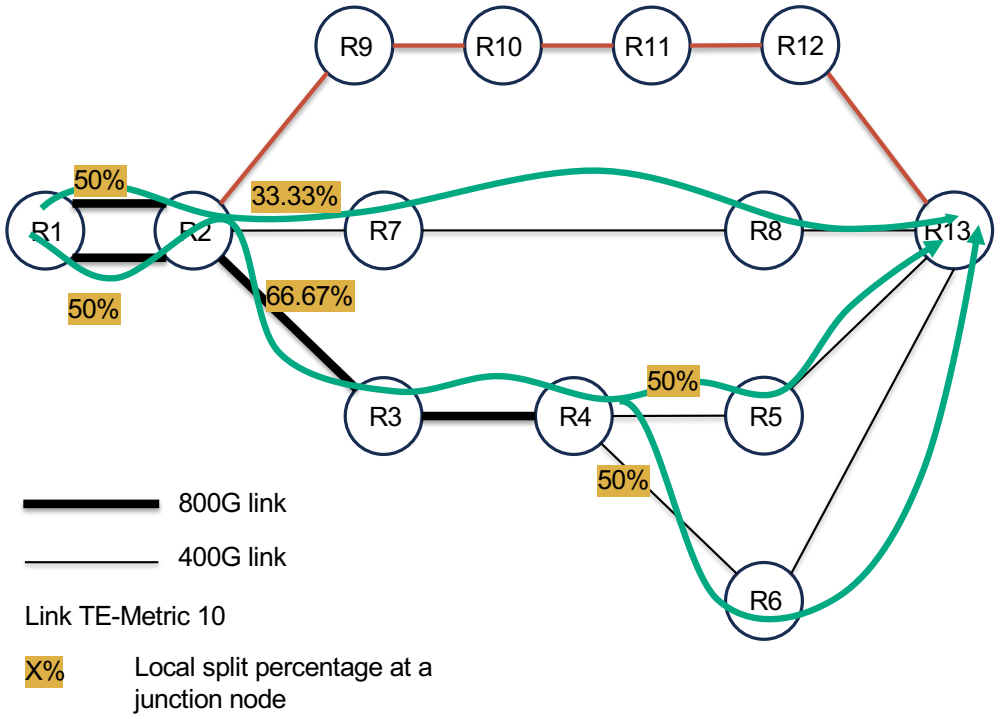
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MPTE DAG – Link Failure



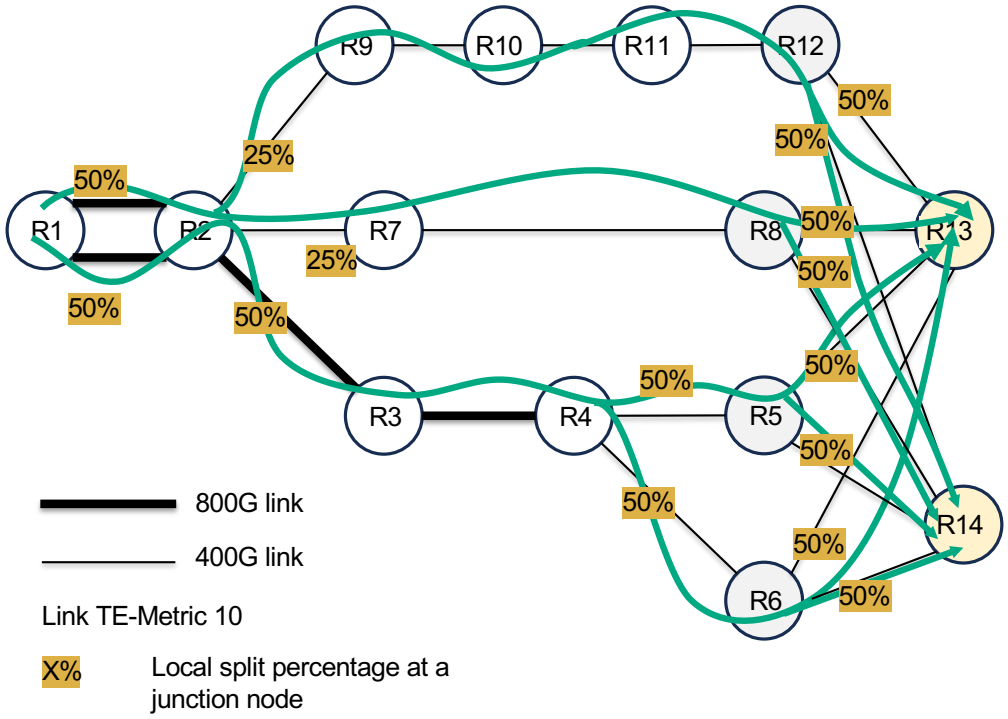
- Automatic adjustment of the MPTE DAG and its branching ratios when a link on the DAG goes down temporarily
- Shape of the DAG remains unchanged unless there is a permanent change to the topology (reduces DAG churn)

MPTE DAG – Constraints



- Regular TE constraints like include/exclude resource affinities apply
  - E.g. Exclude links with red resource affinity

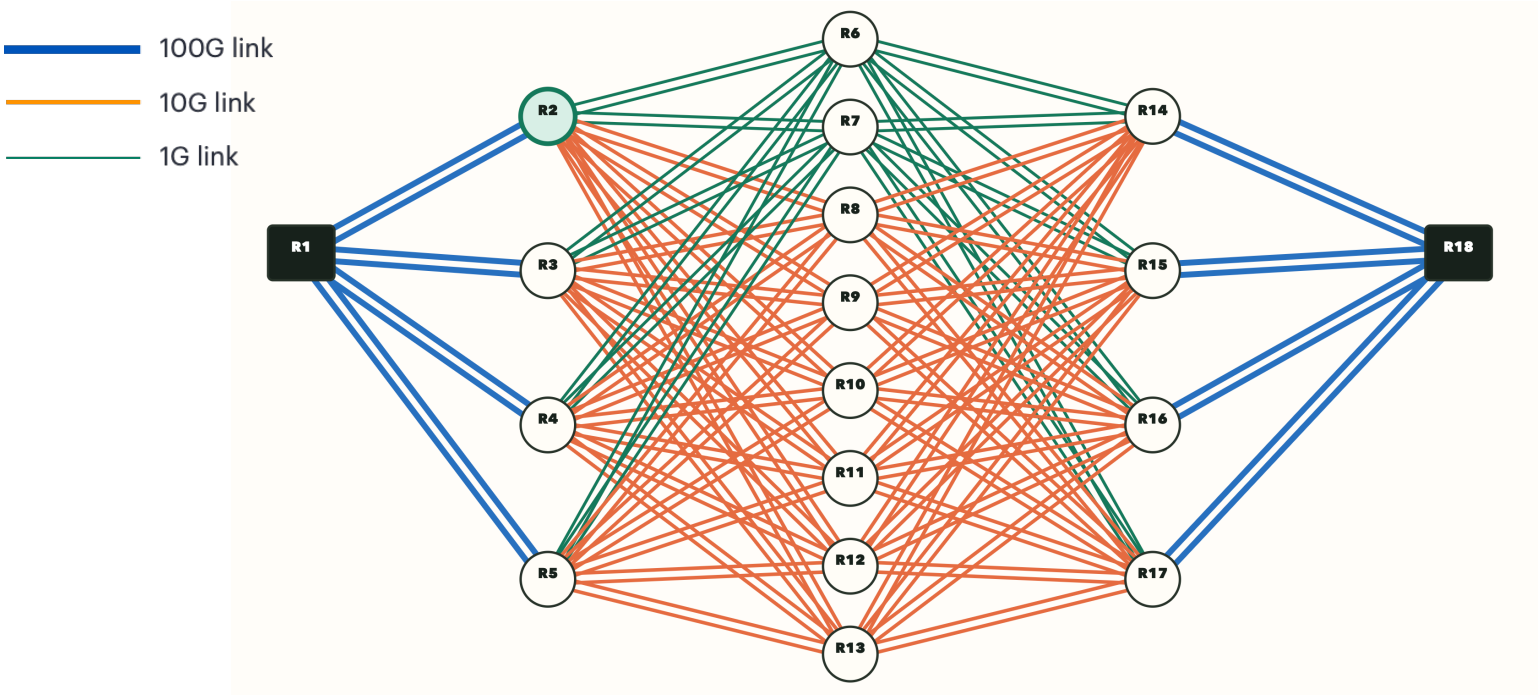
MPTE DAG – Multiple Egresses



- Can have multiple egress PEs for the same MPTE DAG!
  - Note: this is not a multicast P2MP LSP: traffic is load-balanced across the egress PEs.
- R14 can be accommodated by the MPTE DAG with relatively little extra state
  - Only R12, R8, R5 and R6 have updated state; rest of DAG is unchanged

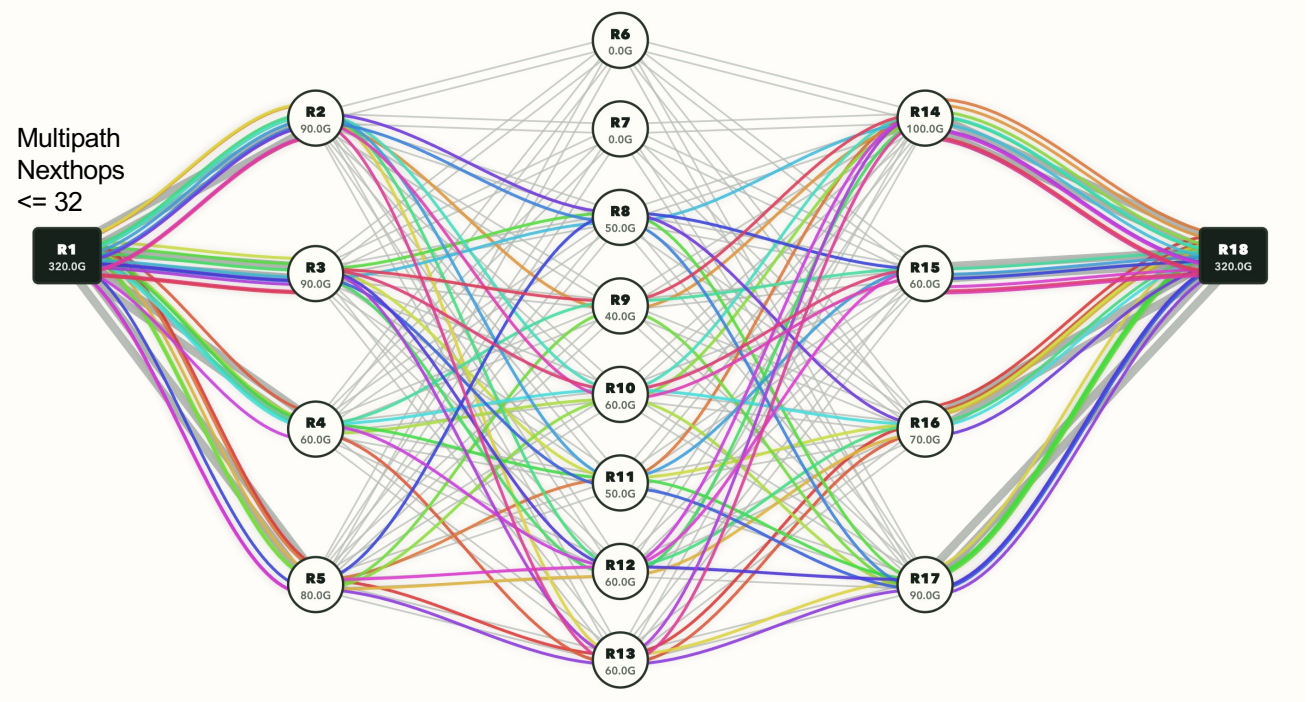
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Example Topology 2



Maximum flow throughput from R1 to R18: 496 Gbps

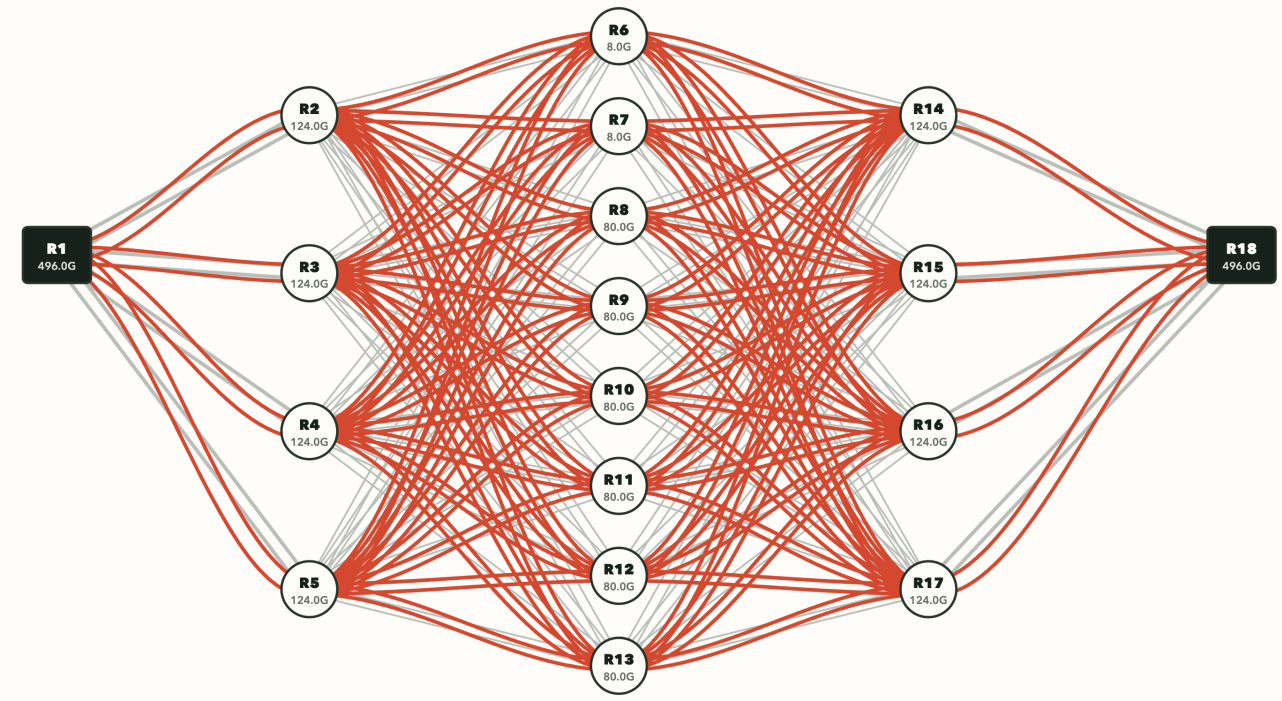
Multipathing using Conventional TE [2]



- With ingress limitation: maximum 32 next hops
  - Requires 32 TE tunnels
  - Maximum achievable throughput: 320 Gbps
- With no ingress next-hop limitation:
  - Requires 64 TE tunnels to realize the full 496 Gbps
    - 48 tunnels over 10 Gbps links

      16 tunnels over 1 Gbps links

Multipathing using Directed Acyclic Graph (DAG) based TE [2]



- MPTE DAG achieves the full 496 Gbps using a single TE tunnel construct
  - Uses 8 next hops at ingress
  - Uses up to 16 next hops at transit
  - Transit nodes R2-R5 distribute traffic unevenly across the 1G and 10G links to maximize throughput

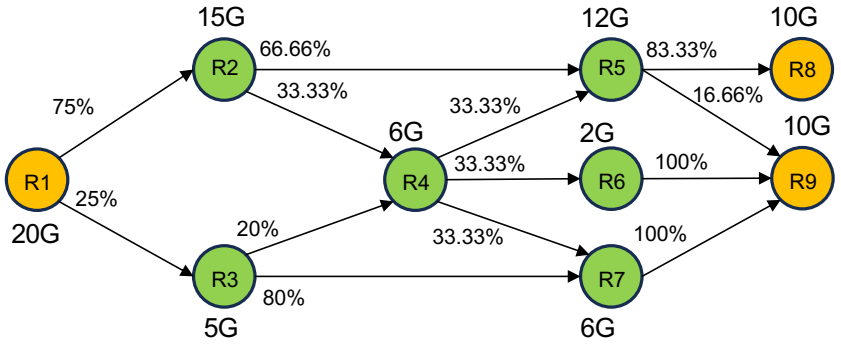
DAG-based MPTE – Architecture

The MPTE DAG is defined by an ingress set, an egress set, an optimization objective, optional slack, and a set of constraints.

- The slack parameter is leveraged to increase the span of the DAG.
- Paradigm shift in computation from a single constrained path to a constrained multipath DAG between ingress and egress sets.

Each participating node is modeled as a junction that maintains a set of previous hops (PHOPs), next hops (NHOPs), bandwidth information, and load-sharing weights.

- This turns the TE object into a per-node forwarding policy across a constrained multipath structure.



Key Differentiators

- Facilitates unequal-cost load balancing at every junction on the DAG – not just on ingress
- Supports multiple ingresses and multiple egresses
- Amount of state needed to setup the tunnel is significantly less
- Amount of churn after a failure event is significantly less

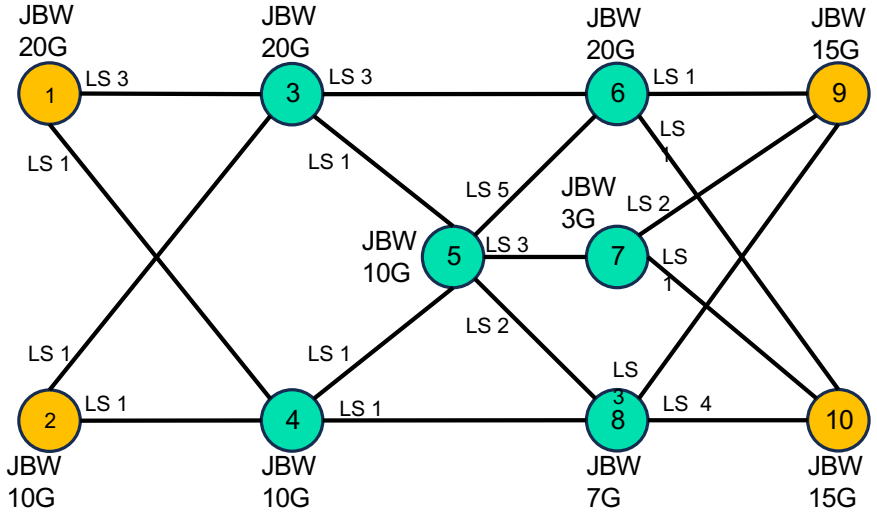
MPTE is transport agnostic and can be applied to **MPLS, Native IP, and SRV6 forwarding models.**

MPTED Tunnels and Junctions

- **MPTED tunnel**
  - TE construct that contains a constrained set of paths representing an optimized Directed Acyclic Graph (DAG) from one or more ingresses to one or more egresses
  - The paths that make up an MPTED tunnel traverse a set of junction nodes
- **MPTED junction**
  - TE construct associated with the MPTED tunnel at each junction node and constitutes a set of previous-hops (JCT-PHOPs) and a set of next-hops (JCT-NHOPs) over which traffic is load-balanced in a weighted fashion

MPTED Tunnel: Tun-West-To-East [30G]

- Ingresses – {1 [20G], 2 [10G]}; Egresses – {9, 10}
- Constraint: Include Green (Resource Affinity)
- Optimization Objective: TE metric



JBW: Junction Bandwidth *LS: Load Share*

- **Provisioning**
  - Provisioning an MPTED tunnel involves provisioning the control and forwarding plane state associated with the MPTED junction at each junction node
  - **Provisioning options: RSVP / BGP / PCEP / gRPC**

Theory Of Operation

MPTED Tunnel Originator (TO)

User specifies *intent*:
- Ingresses
- Egresses
- Incoming bandwidth at each ingress
- Constraints and Optimization Objective



MPTED Computer (MC)

MC computes the DAG:
- Takes the specified constraints and optimization objective into account
- Computation result is a set of unordered list of junctions



MPTED Signaling Source (SS)

SS provisions the MPTED junction state:
- Sends signaling messages to each junction node
- When all junctions are provisioned, ingresses can start sending traffic into the MPTED tunnel

Provisioning Options

	Options
Topology Acquisition	IGP/BGP-LS
Signaling	**RSVP-TE** / PCE / **BGP (new AFI/SAFI)** / YANG based API
Data Plane	**MPLS** or Native IP or SRV6

MPTE RSVP Signaling

Optimized Signaling Procedures

Minimize “Refresh” message processing

Avoid unnecessary signaling adj failures

Minimize “Trigger” message processing

Minimize the number of signaling notifications triggered when a link fails/degrades

Maintain Ordered Control/Programming

(Signaling) Source to Junction (S2J) Messages

- JunctionCreate - RSVP MPTED Path
- JunctionUpdate - RSVP MPTED Path
- JunctionDelete - RSVP MPTED PathTear (with or without CONDITIONS object)

Junction to Source (J2S) Messages

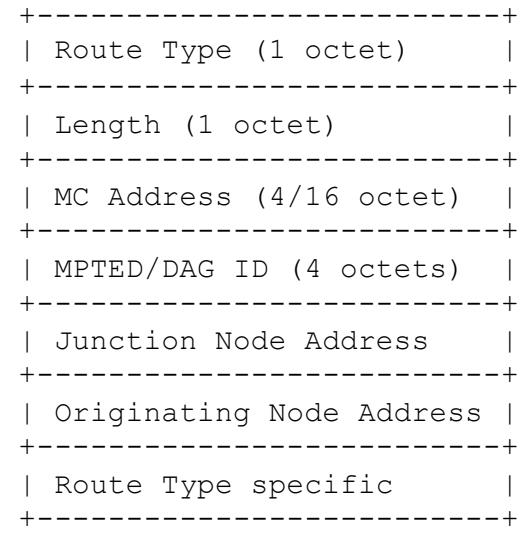
- JunctionNotify - RSVP MPTED Notify
- ResourceNotify - RSVP Rsrc Notify

Junction to Junction (J2J) Messages

- Upstream (J2JU) Messages
  - JunctionNextHopReservation - RSVP MPTED Resv
  - JunctionDown - RSVP MPTED Notify
- Downstream (J2JD) Messages
  - JunctionDelete (Conditional) - RSVP MPTED PathTear (with CONDITIONS object)
  - JunctionNotReady - RSVP MPTED ResvErr

MPTE BGP Signaling

- A New MPTE SAFI
- BGP signaling matches RSVP signaling
  - S2J M-Path –> Junction State Route (Type 1)
  - J2S JunctionNotify -> Junction State Ack Route (Type 2)
  - J2J M-Resv -> Junction Resv Route (Type 3)
    - Omitted for Global-label mode
- PHOP/NHOP Encoded in Tunnel Encapsulation Attribute (TEA)
  - The TEA encodes a list of Tunnel TLVs, each with its own type
  - Each “tunnel” is a PHOP/NHOP, with sub-TLVs encoding specific information
    - Label/BW sub-TLV, etc.; Upstream sub-TLV indicates this is a PHOP
  - The “Any-Encapsulation” tunnel type means any tunnel encapsulation is fine
    - Direct connection, MPLS/GRE/whatever tunnel to the downstream junction



Topology Acquisition in a BGP-Only network

- Typically, IGP floods link state and TE information, and BGP-LS exports that from IGP border routers to a controller
- With BGP-LS-Direct, each BGP speaker advertises its own/direct link state and TE information
  - eBGP peering serves the IGP Hello purposes
- BGP-MPTE and BGP-LS-Direct enable IGP/RSVP-free MPTE in a BGP-only DC
  - IP data plane will be supported as well

MPTE Configuration

```
root@R1_re# set protocols mpted tunnel 1to9 ?
Possible completions:
> admin-group           Administrative group policy
> admin-group-extended  Extended administrative group policy
  compute-engine        Compute engine address
  compute-only          Compute DAG only, do not signal tunnel
  description           Text description of tunnel
  disable               Disable tunnel
+ egress                Tunnel egress address
> hop                   Hop constraints configuration
> ingress               Tunnel ingress configuration
  max-jcts              Maximum number of junctions (1..255)
  max-nhops             Maximum number of nexthops per junction (1..255)
  min-nhops             Minimum number of nexthops per junction (1..255)
> optimize              Optimization configuration
  originator            Tunnel originator address
  priority              Preemption priorities
  signaling-source      Signaling source address
  signaling-type        Signaling protocol type
  type                  Tunnel type
```

Per hop min-bandwidth

```
root@R1_re# set protocols mpted tunnel 1to9 hop ?
Possible completions:
  minimum-bandwidth    Minimum bandwidth per hop (bps)
```

Control the span of the DAG

Multiple optimization objectives

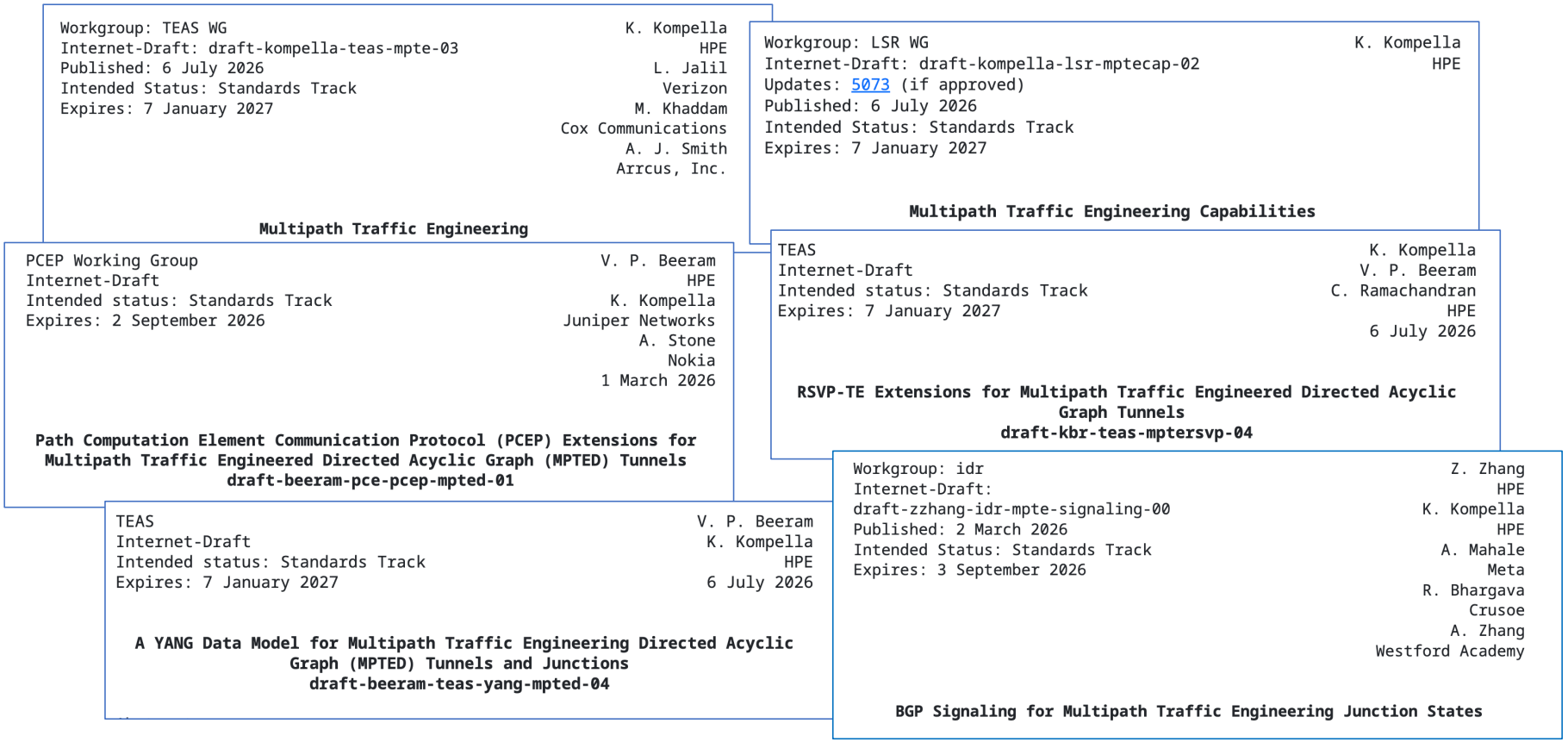
```
root@R1_re# set protocols mpted tunnel 1to9 optimize ?
Possible completions:
  delay                Delay metric optimization
  igp                  IGP metric optimization
  metric-margin        Metric margin for path selection (0..4294967295)
  te                   Traffic engineering metric optimization
```

MPTE – Operational State

```
root@VMX_R1_re> show mpte tunnel extensive
test-tun-1, Originator: 1.1.1.1, ID: 1
  Up, Availability: 100%, Type: MPLS-Sig
  ComputeEngine: 1.1.1.1
  SignalingSource: 1.1.1.1, SignalingType: RSVP
  Ingresses (1):
    1.1.1.1, BW 500Mbps
  Egresses (1):
    2.2.2.2
  OptimizationObjective: TE, Slack: Infinite
  Metric Min 10 Max 30
  DAGVersion 1
    Priorities: 7 7
    Junctions (4):
      JCT 1.1.1.1, Up, ReqBW 500Mbps, MTU 0
        NHOPs (2):
          NHOP 30.1.1.2 Index 358 Metric 20
            HopVersion 1, Up, OutLabel: 299776
              LoadShare 50%, ReqBW 250Mbps, ResBW 250Mbps
          NHOP 20.1.1.2 Index 335 Metric 10
            HopVersion 1, Up, OutLabel: 3
              LoadShare 50%, ReqBW 250Mbps, ResBW 250Mbps
      JCT 2.2.2.2, Up, ReqBW 500Mbps, MTU 0
        PHOPs (2):
        Metric Min 10 Max 30
          PHOP 20.1.1.1 Index 335
            HopVersion 1, Up, InLabel: 3
          PHOP 40.1.1.2 Index 358
            HopVersion 1, Up, InLabel: 3
...
```

```
...
      JCT 3.3.3.3, Up, ReqBW 250Mbps, MTU 0
        PHOPs (1):
        Metric Min 10 Max 10
          PHOP 30.1.1.1 Index 336
            HopVersion 1, Up, InLabel: 299776
        NHOPs (1):
          NHOP 60.1.1.2 Index 359 Metric 20
            HopVersion 1, Up, OutLabel: 299776
              LoadShare 100%, ReqBW 250Mbps, ResBW 250Mbps
      JCT 4.4.4.4, Up, ReqBW 250Mbps, MTU 0
        PHOPs (1):
        Metric Min 20 Max 20
          PHOP 60.1.1.1 Index 359
            HopVersion 1, Up, InLabel: 299776
        NHOPs (1):
          NHOP 40.1.1.1 Index 336 Metric 10
            HopVersion 1, Up, OutLabel: 3
              LoadShare 100%, ReqBW 250Mbps, ResBW 250Mbps
    8 Sep 22 20:08:48.145 Up - 100%
    7 Sep 22 20:08:48.144 Junction Up - 3.3.3.3
    6 Sep 22 20:08:48.144 Junction Up - 4.4.4.4
    5 Sep 22 20:08:48.143 Junction Up - Local 1.1.1.1
    4 Sep 22 20:08:48.143 Junction Up - 2.2.2.2
    3 Sep 22 20:08:48.132 DAG: Computation Result Accepted - 4
junctions, 4 TE links
    2 Sep 22 20:08:48.132 Originate DAG Version - 1
    1 Sep 22 20:08:48.132 Tunnel Created
  Created: Tue Sep 22 20:08:48 2026
```

MPTE References – IETF Drafts



Workgroup: TEAS WG
Internet-Draft: draft-kompella-teas-mpte-03
Published: 6 July 2026
Intended Status: Standards Track
Expires: 7 January 2027

K. Kompella
HPE
L. Jalil
Verizon
M. Khaddam
Cox Communications
A. J. Smith
Arrcus, Inc.

Multipath Traffic Engineering

Workgroup: LSR WG
Internet-Draft: draft-kompella-lsr-mptecap-02
Updates: 5073 (if approved)
Published: 6 July 2026
Intended Status: Standards Track
Expires: 7 January 2027

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Multipath Traffic Engineering Capabilities

PCEP Working Group
Internet-Draft
Intended status: Standards Track
Expires: 2 September 2026

V. P. Beeram
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Juniper Networks
A. Stone
Nokia
1 March 2026

Path Computation Element Communication Protocol (PCEP) Extensions for Multipath Traffic Engineered Directed Acyclic Graph (MPTED) Tunnels
draft-beeram-pce-pcep-mpted-01

TEAS
Internet-Draft
Intended status: Standards Track
Expires: 7 January 2027

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6 July 2026

RSVP-TE Extensions for Multipath Traffic Engineered Directed Acyclic Graph Tunnels
draft-kbr-teas-mptersvp-04

TEAS
Internet-Draft
Intended status: Standards Track
Expires: 7 January 2027

V. P. Beeram
K. Kompella
HPE
6 July 2026

A YANG Data Model for Multipath Traffic Engineering Directed Acyclic Graph (MPTED) Tunnels and Junctions
draft-beeram-teas-yang-mpted-04

Workgroup: idr
Internet-Draft:
draft-zzhang-idr-mpte-signaling-00
Published: 2 March 2026
Intended Status: Standards Track
Expires: 3 September 2026

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Crusoe
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Westford Academy

BGP Signaling for Multipath Traffic Engineering Junction States

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Workshop Coverage

- MPTE Basics
- RSVP and BGP signaling
- Configuration
  - Base IGP/BGP/TE/Overlay configuration (provided)
  - Sample MPTE Configuration (provided)
    - Participants may configure additional MPTE tunnels
- Monitoring & Visualization
  - Checking tunnel and junction states
  - Checking forwarding states
  - Topology Changes
- Advanced Topics

Workshop Topology

- Two CLOS AIDCs connected by a WAN DCI
  - Two border leaves in each AIDC as the DCI PEs
  - BGP signaling in AIDC, ISIS/RSVP in DCI
  - L3VPN over DCI, EVPN over DC
- Each router/switch is a cRPD instance
  - With kernel-based IP/MPLS forwarding
  - All 30 cRPD instances in the same Ubuntu 22.04 VM

Workshop - Activity Sheet

- Visualize and get familiarized with the topology
- Set up multiple MPTE DAG tunnels
  - Use RSVP over an MPLS Forwarding Plane with Distributed Label Assignment in the DCI; Use BGP in the DC
  - Use different optimization objectives
  - Use different metric-margin variations (infinite, zero, non-zero)
  - Use different TE constraints (static-bandwidth, affinities, explicit topological-element exclusions)
  - Use constraints that control the span of the DAG
- Observe and visualize the provisioned TE DAGs, the corresponding programmed routes, and the flow of the traffic
  - Leverage MCP server to fetch nuanced insights
- Trigger scenarios which result in the tunnels to get dynamically updated (in-place or otherwise)
- Delete tunnels



Thank you

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